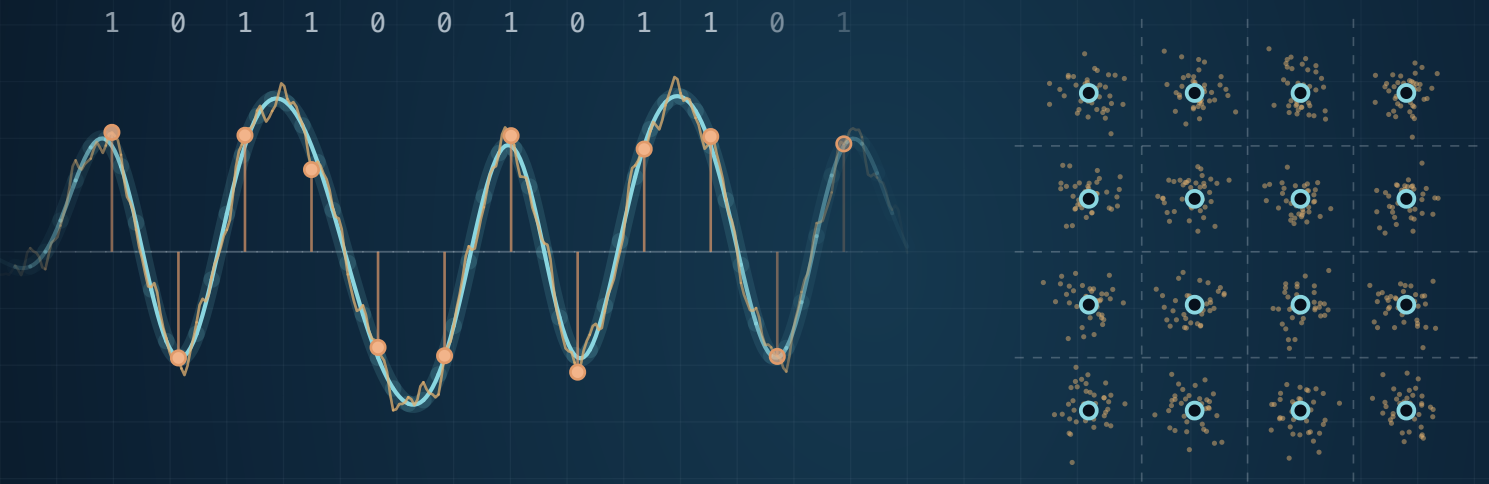




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Laboratory Sheets



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Contents

The course has four laboratories, and each one takes a result the chapters derive on paper and asks you to measure it instead. These sheets run all four. Every curve, cloud and staircase on the pages that follow was computed when this page was drawn. The noise was generated, the filter was applied, the errors were counted. So what you see is the outcome of a run and not a drawing of one.

That matters for one reason above all. A formula tells you what should happen. A measurement tells you what did. Where the two agree you have learnt that the derivation is sound. Where they disagree you have learnt something better: which assumption in the derivation the experiment broke. Each of the four laboratories below ends on that disagreement, because it is the part of the laboratory worth carrying away.

	List of Figures	3
	List of Tables	4
1	Quantization, the signal-to-quantization-noise ratio and PCM	5
	Build a uniform quantizer, measure the SQNR, and encode the samples as a polar-NRZ waveform.	
2	The matched filter and the bit error rate	8
	Measure the bit error rate of a matched-filter receiver and compare it with the Chapter 2 formula.	
3	Sixteen-point QAM and the union bound	10
	Measure the symbol error rate of sixteen-point QAM and compare it with the Chapter 4 bound.	
4	Huffman coding and how much it compresses	12
	Build a Huffman code and measure its compression.	

How to use the sheets

Your numerical results will differ from these examples. Each group receives a different input number, so its waveform, bit pattern, and source statistics differ. Random noise also changes between runs. Compare curve shapes, gaps, trends, and expected directions instead of individual sample values.

Chapter and section numbers, such as Chapter 2 or Section 1.4, refer to the Lecture Notes.

List of Figures

LABORATORY 1 · QUANTIZATION, THE SIGNAL-TO-QUANTIZATION-NOISE RATIO AND PCM

1.1	The first second of the run.	5
1.2	The quantizer itself, drawn by feeding it a thousand values across its range.	6
1.3	The first four samples encoded and sent.	6
1.4	Measured against modelled, for one to six bits a sample, both computed from the seventeen samples the laboratory takes.	6

LABORATORY 2 · THE MATCHED FILTER AND THE BIT ERROR RATE

2.1	Sixteen bits at $E_b/N_0 = 2$ dB.	8
2.2	The matched-filter output for the same sixteen bits, read once at the end of every bit interval.	8
2.3	Measured bit error rate against the formula, five thousand bits a point.	9

LABORATORY 3 · SIXTEEN-POINT QAM AND THE UNION BOUND

3.1	$E_b/N_0 = 6$ dB.	10
3.2	$E_b/N_0 = 11$ dB.	10
3.3	Measured symbol error rate against the nearest-neighbour approximation.	11

LABORATORY 4 · HUFFMAN CODING AND HOW MUCH IT COMPRESSES

4.1	The source statistics p_i over the ten-digit alphabet.	12
4.2	The code the algorithm builds.	12
4.3	$\rho = 0.01$	13
4.4	$\rho = 0.99$	13

List of Tables

LABORATORY 4 · HUFFMAN CODING AND HOW MUCH IT COMPRESSES

4.1 Entropy, average codeword length, efficiency and compression for the three sources. 13

LABORATORY

1

Quantization, the signal-to-quantization-noise ratio and PCM

The first laboratory applies the complete Chapter 1 chain. Start with a continuous waveform and sample it at the Nyquist rate. Quantize the samples with a uniform quantizer. Measure the ratio of signal power to quantization-error power. Encode each sample as a binary word and draw the polar-NRZ waveform. Then double the number of levels and repeat the measurements.

The waveform is $x(t) = f_1 \sin(2\pi f_1 t) + f_2 \cos(2\pi f_2 t)$. Each group receives integer values for f_1 and f_2 . Here, $f_1 = 3$ and $f_2 = 4$, so the Nyquist rate is 8 samples per second. A two-second interval then contains only seventeen samples.

The quantizer itself is four lines. Find the smallest and largest sample. Divide the distance between them into L equal cells. Therefore, the step is $\Delta = (x_{\max} - x_{\min})/L$. Decide which cell each sample falls in. Replace it by the middle of that cell. This is the value that minimises the error inside the cell when nothing is known about where in the cell the sample sits.

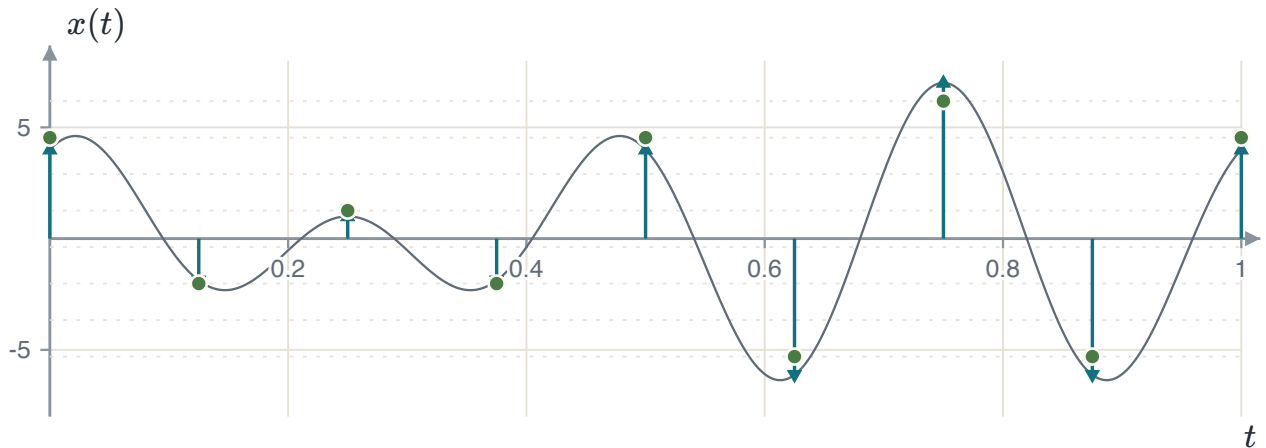


FIGURE 1.1 The first second of the run. The pale line is $x(t)$. Stems show samples x_n taken every $1/8$ s. Filled circles show quantized values $\hat{x}_n$ for $L = 8$. Dotted lines show the eight representation levels with spacing $\Delta = 1.640$ V.

With $L = 8$ each sample needs $R = \log_2 8 = 3$ bits, and at $f_s = 8$ samples a second the stream leaves the encoder at $R_b = Rf_s = 24$ bit/s. At $L = 16$ it is 32 bit/s. That is the price of the second experiment, and it is worth naming before the reward is measured: a third more bits a second.

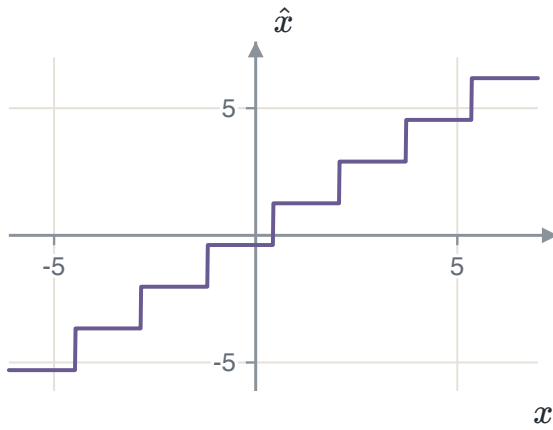


FIGURE 1.2 The quantizer itself, drawn by feeding it a thousand values across its range. Eight treads, each $\Delta = 1.640$ V wide, each one tread high.

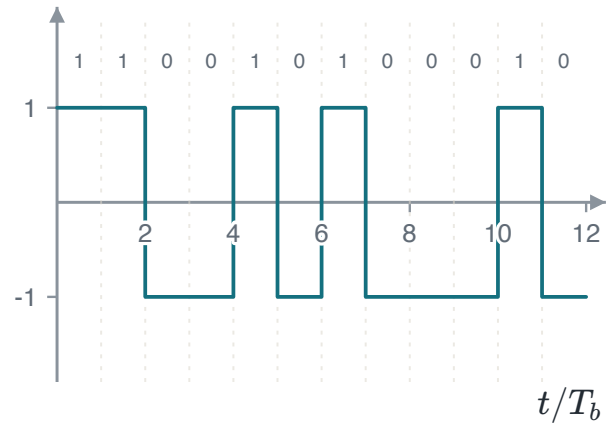


FIGURE 1.3 The first four samples encoded and sent. Each sample becomes a three-bit word for its cell, and each bit becomes one bit interval of a polar NRZ waveform. High for a one, low for a zero.

WHAT THE LABORATORY MEASURES

$$\text{SQNR} = \frac{E[X^2]}{E[(X - \hat{X})^2]} \longrightarrow \text{SQNR}|_{\text{dB}} = 10 \log_{10} \text{SQNR}$$

Both averages use the available samples. The numerator is the sample mean-square value. The denominator is the mean-square quantization error. Chapter 1 models this denominator as $\Delta^2/12$. Here, it is measured directly.

On this waveform the measurement gives 17.80 dB at $L = 8$ and 23.83 dB at $L = 16$. The difference is 6.03 dB. This is the six decibels a bit that Section 1.4 predicts, arrived at by counting. That agreement is the main result of the laboratory, and it holds for a reason that has nothing to do with this particular waveform. Doubling L halves Δ , halving Δ quarters $\Delta^2/12$, and a factor of four is 6.02 dB.

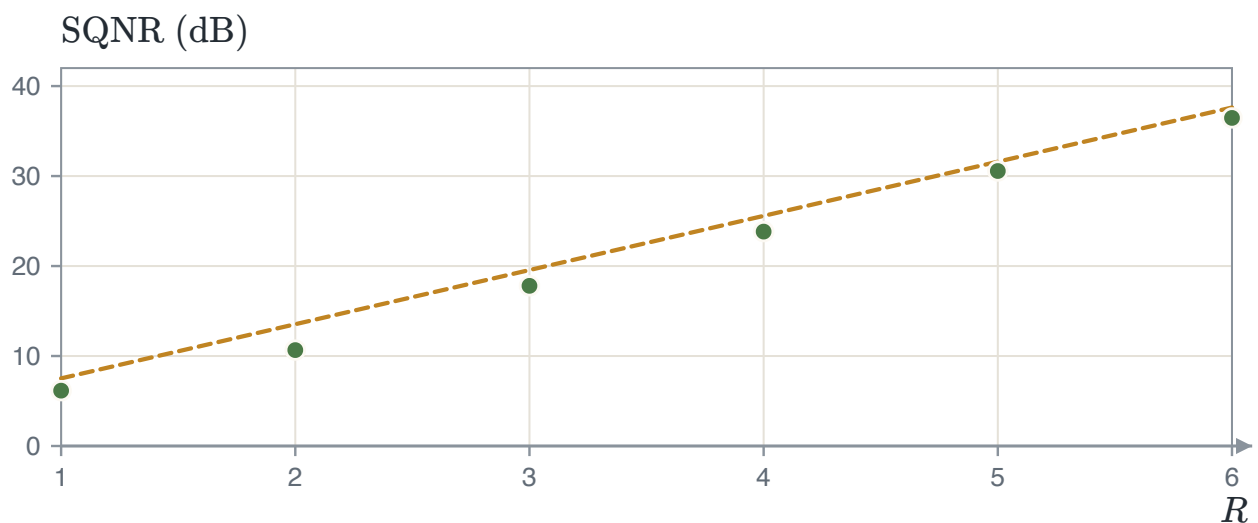


FIGURE 1.4 Measured against modelled, for one to six bits a sample, both computed from the seventeen samples the laboratory takes. The dashed line is the small-step model $10 \log_{10}(P_X/(\Delta^2/12))$ and the circles are the ratio actually measured. The circles rise about six decibels a bit, as they should. But they sit below the line by between 1.0 and 2.9 dB, and the gap neither closes nor settles as R grows. Something is wrong, and the next note says what.

⊗ SEVENTEEN SAMPLES IS NOT A POPULATION

The measured points and the modelled line disagree by up to 2.9 dB, and the gap does not shrink as L grows. It is tempting to blame the model, and Section 1.4 gives a real reason it could be blamed. The error is not quite uniform across a cell. But measure the same waveform with two thousand samples instead of seventeen and the gap falls to less than 0.6 dB everywhere. The model was right. The estimate was noisy. Both $E[X^2]$ and $E[(X - \hat{X})^2]$ are averages, and an average over seventeen numbers has a spread of its own. This is the error to name in your report. Do not draw a conclusion about a model from a sample too small to test it.

⚠ SAMPLING EXACTLY AT THE NYQUIST RATE

The waveform here has a 4 Hz component and the laboratory samples at exactly 8 Hz. This is the boundary the sampling theorem is stated at rather than inside. At exactly twice its frequency, a cosine is sampled at the same two points of every cycle. Therefore, that component contributes the same alternating ± 4 to every sample and none of its shape survives. The reconstruction is still correct here because the component is a cosine and lands on its peaks. Move the phase and it would not be. Chapter 1 asks for $f_s > 2W$ and means the strict inequality.

LABORATORY

2

The matched filter and the bit error rate

The second laboratory builds the Chapter 2 receiver. Convert a bit stream into a waveform and add white Gaussian noise. Apply the matched filter and sample its output once per bit. Compare each sample with the decision threshold. Then compare the measured error rate with $Q(\sqrt{2E_b/N_0})$.

The signaling is antipodal. The pulse $\psi(t) = 1/\sqrt{T_b}$ has unit energy over one bit interval. A bit scales this pulse by $\pm A\sqrt{T_b}$. Thus, the transmitted waveform is constant at $+A$ or $-A$ during each interval. Its bit energy is $E_b = A^2 T_b$. Add sample noise with variance $N_0/2$ and use $10 \log_{10}(E_b/N_0)$.

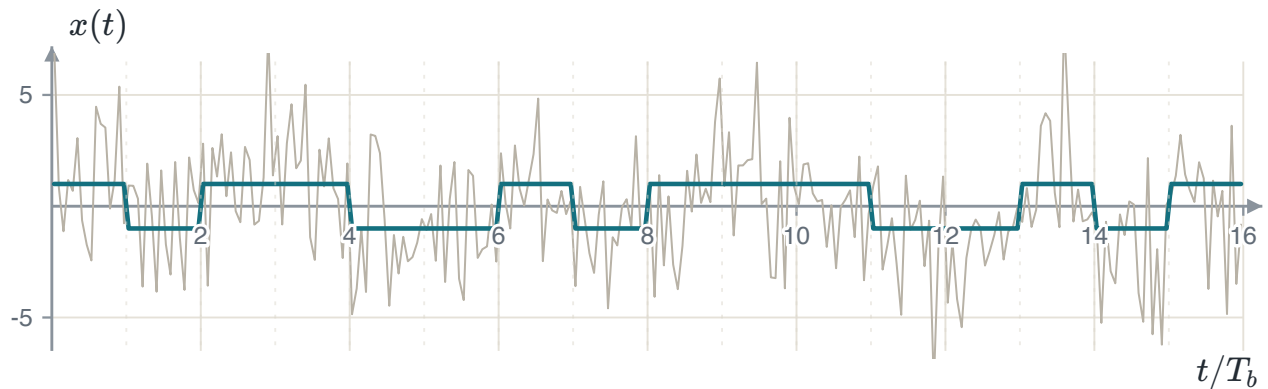


FIGURE 2.1 Sixteen bits at $E_b/N_0 = 2$ dB. The clean square wave is the transmitted $s(t)$. The hairline around it is $x(t) = s(t) + w(t)$. This is everything the receiver is given. Nothing about the square wave is visible in the hairline at this noise level, and that is the honest picture. The receiver does not see the signal, it computes with it.

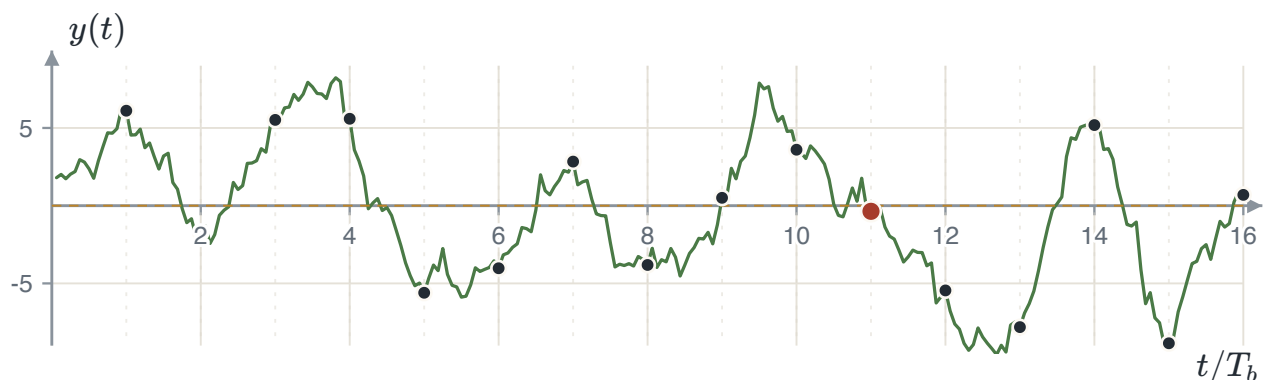


FIGURE 2.2 The matched-filter output for the same sixteen bits, read once at the end of every bit interval. The dashed line is the threshold $\lambda = 0$. This is where equal priors put it. Fifteen of the sixteen readings are on the correct side of it. The eleventh is not, and it is drawn in red. At 2 dB the formula expects about one wrong bit in twenty-seven, and this run produced one in sixteen.

One run of sixteen bits settles nothing. One error could as easily have been none or three. The second part of the laboratory therefore drops the picture and repeats the experiment. Five thousand bits at each signal-to-noise ratio from 0 to 5 dB in quarter-decibel steps, counting errors and dividing by five thousand.

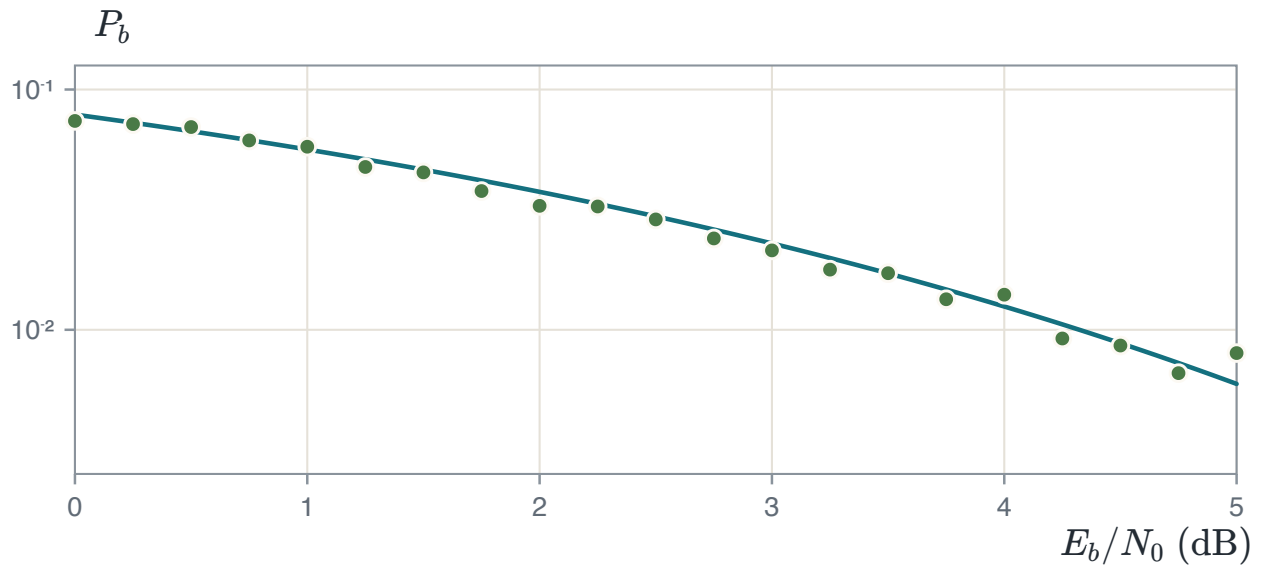


FIGURE 2.3 Measured bit error rate against the formula, five thousand bits a point. The line is $Q(\sqrt{2E_b/N_0})$ and the circles are counted errors. Nowhere on the sweep do the two differ by more than 0.005, and yet the circles clearly wander further from the line towards the right. The next note is about why those two statements are both true.

⚠ MONTE CARLO SCATTER

Counting k errors in N bits gives a standard spread of approximately $\sqrt{p(1-p)/N}$. At 0 dB, $p \approx 0.079$, and five thousand bits give a spread of 0.004. At 5 dB, the absolute spread decreases, but it becomes a larger fraction of p . A logarithmic plot shows this relative scatter. Measuring probabilities near 10^{-6} therefore requires millions of trials.

⊗ THREE-DECIBEL CONVENTION ERRORS

Two convention errors each give a factor of two. First, the sample-noise variance is $N_0/2$ for a two-sided density $N_0/2$. Using N_0 shifts the measured curve by 3 dB. Second, the formula $Q(\sqrt{2E_b/N_0})$ requires antipodal signaling. On-off signaling instead uses $Q(\sqrt{E_b/N_0})$. A parallel 3 dB shift usually indicates one of these errors.

LABORATORY

3

Sixteen-point QAM and the union bound

The third laboratory uses a two-dimensional sixteen-point square constellation. Add independent noise on each axis. The receiver selects the nearest point. Compare the measured symbol error rate with the nearest-neighbor approximation from Section 4.4.

Write each point as $s_I + js_Q$, with coordinates from $\{-3, -1, 1, 3\}$. These units give $E_s = 10$ and $d_{\min} = 2$. Each symbol carries four bits, so $E_b = E_s/4 = 2.5$. Section 5.3 gives $d_{\min}^2 = 6E_s/(M - 1) = 4$. This result agrees with the point spacing.

The same constellation gives the optimal thresholds. Independent axes and equal level spacing produce three vertical and three horizontal boundaries. Each axis uses $\lambda \in \{-2, 0, 2\}$. Thus, nearest-point detection reduces to nearest-level detection on each axis.

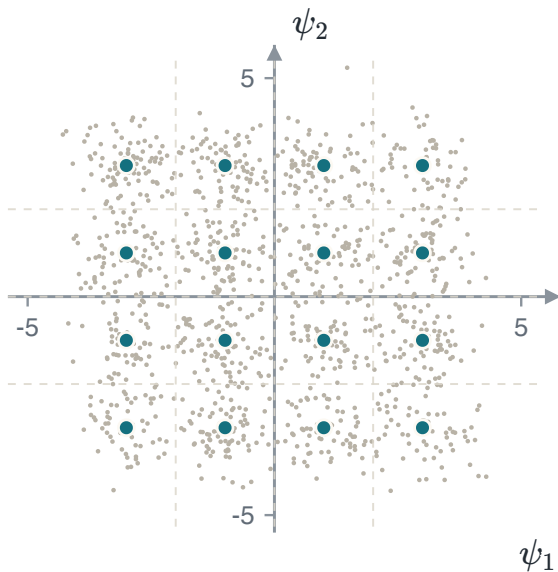


FIGURE 3.1 $E_b/N_0 = 6$ dB. The clouds touch, so a large fraction of the symbols land across a boundary. The measured symbol error rate here is 0.108: about one symbol in nine is decided wrongly.

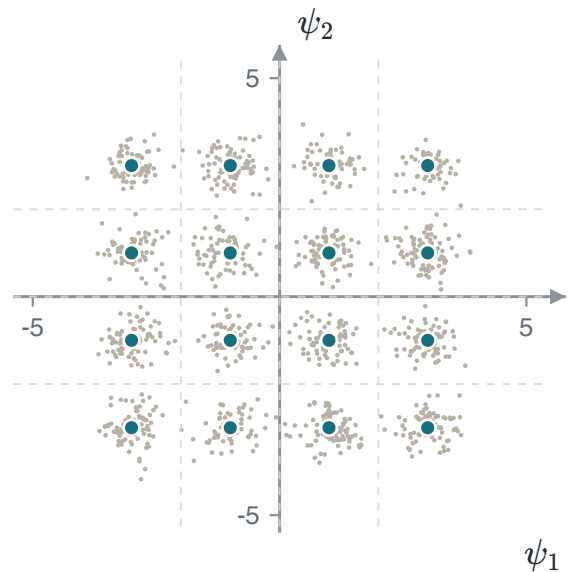


FIGURE 3.2 $E_b/N_0 = 11$ dB. Five decibels more energy a bit, and the clouds have pulled apart into sixteen separate blobs. The measured rate falls to 0.0022, about fifty times better.

Twelve hundred received symbols are drawn in each picture so that the clouds can be seen. The rates quoted beneath them do not come from those twelve hundred. A rate of 0.0022 measured on twelve hundred symbols would be three errors, and three is not a measurement. They come from the sweep below. This uses fifty thousand symbols at 6 dB and a quarter of a million at 11 dB. Choosing how many symbols to run is part of the experiment, and Laboratory 2 said why.

WHAT THE BOUND PREDICTS

$$P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right), \quad N_{\min} = 3, \quad d_{\min}^2 = 4$$

$N_{\min}$ is the average number of nearest neighbours: four corner points have two each, eight edge points three, and four inner points four, and $(4 \cdot 2 + 8 \cdot 3 + 4 \cdot 4)/16 = 3$. It is an average over the constellation and it does not have to be a whole number.

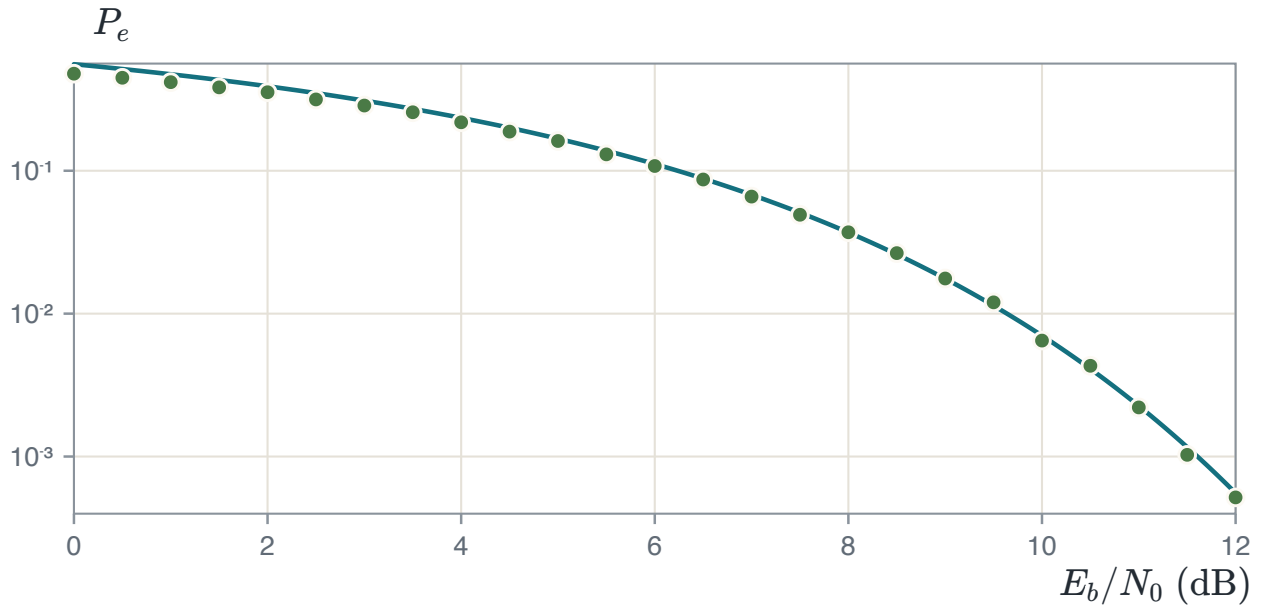


FIGURE 3.3 Measured symbol error rate against the nearest-neighbour approximation. The number of symbols is raised as the rate falls. Fifty thousand at the left, four hundred thousand at the right. So that every circle rests on a few hundred errors. The circles sit below the line at the left and join it by about 8 dB. That gap is not an error in either the simulation or the formula. It is what the approximation is.

✓ MEASURED ERROR AND THE UNION BOUND

The union bound adds all pairwise error probabilities. These events overlap, so the sum can count one noise vector more than once. The bound is therefore larger than the exact union probability. At low signal-to-noise ratio, overlap makes the bound loose. At high ratios, nearest-neighbor errors dominate and overlap becomes small. Remaining differences between simulated points and the bound can come from finite trial counts.

⊗ SYMBOL AND BIT ENERGY

For sixteen-point QAM, $E_s = 4E_b$. Convert between these energies once when calculating N_0 . Do not calculate N_0 from one energy and use the other in the Q function. This error shifts the complete curve by $10 \log_{10} 4 = 6.02$ dB.

LABORATORY

4

Huffman coding and how much it compresses

The fourth laboratory studies lossless source coding. A source emits symbols with known probabilities. Entropy gives the minimum average bits per symbol. Huffman coding gives the best single-symbol prefix code. The laboratory measures both values.

The source alphabet is the ten digits. Your group counts how often each digit occurs in a number it is given and divides by how many digits there are. This gives a probability for each. The case run here uses the eleven digits 1 2 3 4 5 6 5 4 3 2 1. Five digits occur twice and one occurs once. Therefore, five symbols have probability $2/11$, one has $1/11$, and four have probability zero and take no codeword at all.

⊗ DIVIDING BY THE WRONG TOTAL

The probabilities must come from dividing each count by the *number of symbols*. Dividing by anything else. The sum of the digit values, say. Gives a set of numbers that does not add up to one, and everything downstream is then quietly wrong. The entropy is not an entropy, the codebook is built for a source that does not exist, and the efficiency can come out above 100%. Print the sum of your probability vector and confirm it is one before you compute anything from it. It is one line, and it catches the only mistake in this laboratory that produces plausible numbers instead of an error message.

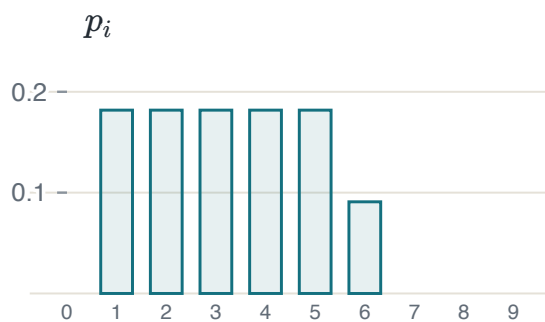


FIGURE 4.1 The source statistics p_i over the ten-digit alphabet. Four digits do not occur, and a symbol of probability zero is left out of both the entropy and the codebook.

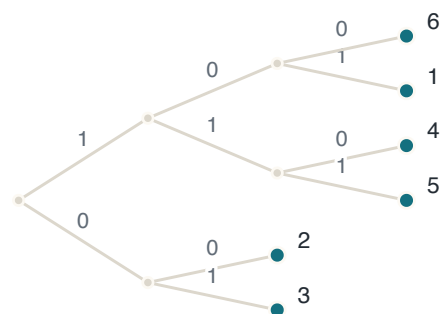


FIGURE 4.2 The code the algorithm builds. Reading a path from the root, left is a 0 and right a 1. The digit at each leaf is the symbol that path spells.

This source has entropy $H = 2.550$ bits per symbol. Its Huffman code has average length $\bar{L} = 2.636$, so $\eta = 96.7\%$. First, check the Section 6.2 bound $H \leq \bar{L} < H + 1$. Here, $2.550 \leq 2.636 < 3.550$. Second, compare $\bar{L}$ with the three-bit fixed length for six symbols and the four-bit length for ten symbols.

⚠ A TIE IN THE ALGORITHM IS NOT A MISTAKE

Whenever two nodes have the same probability, the algorithm may merge either, and different choices give different codebooks with the same average length. Your codewords will not match the ones drawn above and need not. What must match is $\bar{L}$, because Huffman is optimal and the optimal average length is unique even when the code is not. If two groups get different average lengths for the same statistics, one of them has an error. If they get different codebooks, neither of them does. Section 6.3 develops the tie-breaking rule that also minimises the variance of the codeword length.

The second part encodes one thousand symbols from the same distribution. Four-bit fixed-length coding uses 4000 bits. Huffman coding uses 2621 bits, so the measured compression is 34.5%. The average length predicts $1 - \bar{L}/4 = 34.1\%$. These values need not match exactly because the finite sequence is one random sample. A large difference can indicate an incorrect codebook.

The third part asks what the source statistics themselves are worth. Ten counts are drawn from a binomial trial with 50 attempts and success probability ρ , then normalised into a distribution. At $\rho = 0.01$ almost every attempt fails, so most counts come out zero and the surviving symbols are few. At $\rho = 0.99$ almost every attempt succeeds, so all ten counts come out near 50 and the distribution is nearly flat.

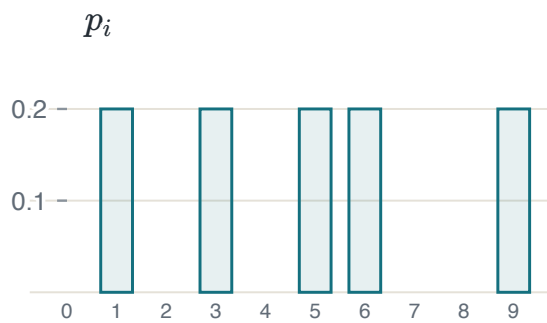


FIGURE 4.3 $\rho = 0.01$. Five symbols survive and the rest have probability zero. $H = 2.322$ bits, $\bar{L} = 2.400$ bits.

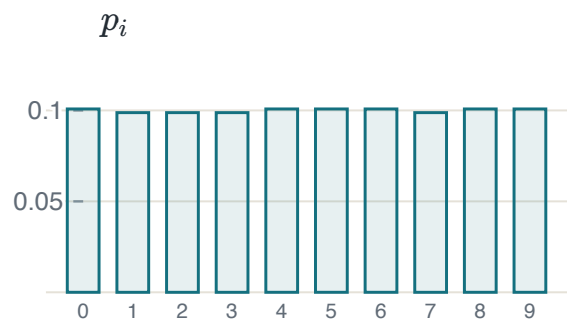


FIGURE 4.4 $\rho = 0.99$. All ten symbols occur about equally often. $H = 3.322$ bits, $\bar{L} = 3.395$ bits.

TABLE 4.1 Entropy, average codeword length, efficiency and compression for the three sources.

SOURCE	H	$\bar{L}$	η	COMPRESSION AGAINST 4 BITS
The digit source	2.550	2.636	96.7%	34.1%
$\rho = 0.01$	2.322	2.400	96.7%	40.0%
$\rho = 0.99$	3.322	3.395	97.8%	15.1%

✓ EFFICIENCY AND COMPRESSION

The coding efficiency changes from 96.7% to 97.8%, while compression decreases from 40% to 15%. Efficiency measures the gap between the code length and the source entropy. Compression compares the code length with a four-bit fixed-length code. At $\rho = 0.99$, the source is nearly uniform and has entropy 3.322 bits. Therefore, little lossless compression remains possible.

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VERSION	DATE	CHANGES
v0	24 September 2026	First edition.

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